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Review

Remediation of soil contaminated with organic compounds by nanoscale zero-valent iron: A review

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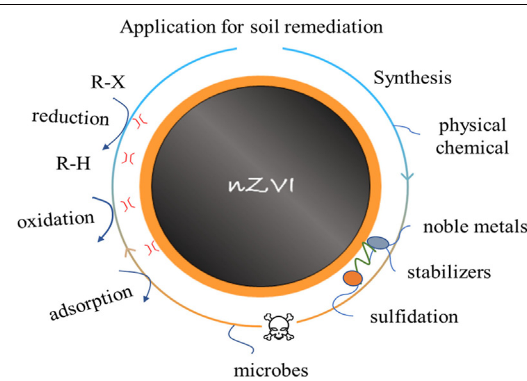
HIGHLIGHTS

- Synthesized by various methods, nZVI is used to degrade organic pollutants.
- Chemical and biological modifications are used to improve the activity of nZVI.
- Electrokinetics, chemical, and microbial methods assist nZVI for soil remediation.

GRAPHICAL ABSTRACT



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ABSTRACT

In recent years, nanoscale zero-valent iron (nZVI) has been gradually applied in soil remediation due to its strong reducing ability and large specific surface area. Compared to conventional remediation solutions, in situ remediation using nZVI offers some unique advantages. In this review, respective merits and demerits of each approach to nZVI synthesis are summarized in detail, particularly the most commonly used aqueous-phase reduction method featuring surface modification. In order to overcome undesired oxidation and agglomeration of fresh nZVI due to its high reactivity, modifications of nZVI have been developed such as doping with transition metals, stabilization using macromolecules or surfactants, and sulfidation. Mechanisms underlying efficient removal of organic pollutants enabled by the modified nZVI lie in alleviative oxidation and agglomeration of nZVI and enhanced electron utilization efficiency. In addition to chemical modification, other assisting methods for further improving nZVI mobility and reactivity, such as electrokinetics and microbial technologies, are evaluated. The effects of different remediation technologies and soil physicochemical properties on remediation performance of nZVI are also summarized. Overall, this review offers an up-to-date comprehensive understanding of nZVI-driven soil remediation from scientific and practical perspectives.

Contents

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1. Introduction

With the fast development of industry, organic compounds have become the world's most widely occurring contaminants in soil, groundwater, and sediments (Rivett et al., 2005). The organic compounds dissolved in water or non-aqueous phase liquids (NAPLs) especially threaten public health as they are not only recalcitrant for transformation and also mobile for transport. Conventional in situ soil and groundwater remediation approaches involved pump-and-treat (Mackay and Cherry, 1989), thermal treatment (Kingston et al., 2010), surfactant/co-solvent flushing (Dugan et al., 2010; Mao et al., 2015), chemical oxidation (Siegrist et al., 2011; Bekele and Naidu, 2015), chemical reduction (Stroo et al., 2012; Wit, 2013) and bioremediation (Kuppusamy et al., 2016). These technologies have been widely implemented for NAPL removal, but their intrinsic drawbacks limited performance and sustainability. These drawbacks lie in relatively high expense for photocatalysts, potential secondary pollution for thermal desorption, and limited efficiency for bioremediation (Siegrist et al., 2011; Stroo et al., 2012; Zhao et al., 2019).

Since it was first synthesized in 1994, nanoscale zero-valent iron (nZVI) has been recognized as an effective material for soil remediation due to its strong reducing ability and large specific surface area (Gillham and Ohannesin, 1994; Zhang et al., 2011). Previous studies suggested that, in situ remediation using nZVI was superior conventional remediation solutions due to several trademark advantages. Various synthesis methods, including chemical hydrothermal synthesis, precision milling, and electrochemical synthesis, have been developed over the past decades (Chen et al., 2004; Li et al., 2009; Woo et al., 2014; Kozma et al., 2016). Particularly, further modification methods have been applied to enhancing reactivities of fresh nZVI for practical soil remediation, such as doping with transition metals, stabilization using macromolecules or surfactants, and sulfidation (Lin et al., 2004; Basnet et al., 2013; Li et al., 2017; Mu et al., 2017). Synergy of nZVI with other technologies also was documented (Koenig et al., 2016). For example, Zanaroli et al. (2012) reported that coupled nZVI and organohalide-respiring bacteria resulted in faster removal of organohalides than nZVI or biodegradation alone. This review aims at presenting the state of the art in the nZVI technology, identifying the current challenges, offering perspectives for future research, and providing a theoretical and practical reference specifically for the application of nZVI in soil remediation. We systematically summarized and critically reviewed the synthesis and modification methods of nZVI, the application of nZVI for in situ organic contaminated soil remediation, and how soil physicochemical properties affected nZVI performance.

2. Synthesis and modification of nZVI

2.1. Synthesis of nZVI

The synthesis methods of nZVI mainly included physical methods, chemical methods and microbial methods according to the synthesis mechanism. Physical methods broke down bulk iron materials to form nanoscale materials by means of mechanical forces and high energy. The prior physical method was severe plastic deformation (SPD), which occurred under quasi-static pressure that resulted in significant refinement of the nanostructure (Valiev et al., 2000). It overcame several difficulties connected with residual porosity and impurities, but the pressure was difficult to control. Vapor deposition (also called evaporative condensation) was later invented, utilizing vacuum evaporation, laser heating evaporation, electron beam irradiation and sputtering to gasify the raw materials and rapidly condense them in the medium (Cheng et al., 2006). With a high energy demand, it was not suitable for large-scale applications. The most widely accepted physical method was precision milling using an agitator to stir up the milling medium (beads) at a designated rotating speed (Li et al., 2009), during which a superfine nanoscale power was formed by solid-state reaction or diffusion. Testing of the milled nZVI (8 h) showed high reactivity towards organic pollutants, which was promising for industrial production. Unfortunately, the irregular shape of produced nZVI and the high energy consumption of the physical methods limit their application.

In contrast to physical methods, chemical methods are more commonly used because of their simplicity and high controllability. The classic aqueous-phase reduction using Fe^{2+} and sodium borohydride ($NaBH_4$) as substrates, without heating or catalysis (Eq. (1)), was the most frequently applied method. Woo et al. (2014) proved that different washing solutions and drying conditions could alter the surface characteristics of nZVI and its reactivity towards reducible contaminants. For example, volatile solvent washing and anaerobic drying conditions enhanced nZVI reactivity towards nitrate. However, the estimated high cost for aqueous-phase reduction ($200/kg) limited its application (Yan et al., 2013). Comparatively, carbothermal reduction was more cost effective (by using less expensive carbon materials as substrates) and reduced Fe^{2+} with gaseous reducing agents (Eqs. (2) and (3)) in which the pyrolytic temperature had a distinct effect on the form of nZVI (>500 °C) (Hoch et al., 2008). An electrolysis-based method for nZVI production using electrodes (cathode and anode), electric current, and Fe^{2+}/Fe^{3+} salts was less expensive and simpler compared to aqueous-phase reduction and carbothermal reduction. The challenge of keeping the formed nZVI from aggregating at the cathode was solved by ultrasonication and ion exchange in Nafion film (Chen et al., 2004; Wang et al., 2008).

Green synthesis of nZVI using plant extracts and microorganisms to reduce Fe^{2+} or Fe^{3+} has gained much attention due to several limitations of the above methods (Kozma et al., 2016). The plant extracts have less toxicity, higher water solubility, and high yield and can synthesize nZVI with longer longevity and varied morphology and structure (round, square, and irregular). The biomineralization-induced approach induced coagulation, nucleation and precipitation of nZVI by changing the physical and chemical conditions of the surrounding microenvironment (Wu, 2017). The biomineralization-inspired approach formed inorganic minerals using the organisms' biochemical processes in vivo (Crespo et al., 2017). The main differences between these two

Table 1
Summary of synthesis methods of nZVI.

Methods		Sizes (nm)	Surface area (m^2/g)	Pros	Cons	References
Physical synthesis	Severe plastic deformation	7–12 or 120–650	N.A.	No residual porosity and impurity	High processing conditions and special equipment	Valiev et al., 2000
	Vapor deposition	3.5–9	N.A.	Small size and high purity	High energy consumption and high cost	Mu et al., 2014
	Precision milling	10–50	39	Superior reactivity, low energy consumption	Limited control over size and morphology	Li et al., 2009
Chemical synthesis	Aqueous-phase reduction	30–100	7.3–68	Easy to operate, high reactivity	Reducing agent is toxic and high-cost	Woo et al., 2014
	Carbothermal reduction	20–150	130	Cheap reducing agent, easy to production	Require high temperature	Hoch et al., 2008
	Electrochemical method	1–20	25.4	High density and small size	Easy to form nZVI clusters	Chen et al., 2004
Green synthesis	Plant extracts	20–120	5.8	Nontoxic reducing agent	Irregular shapes	Kuang et al., 2013
	Microbial synthesis	10–24.6	N.A.	Ecofriendliness, low-cost	Not well known	Tarafdar and Raliya, 2013

microbial approaches were whether the shape, size, and structure of the resulting minerals were precisely and strictly controlled by the organisms. The pros and cons of these methods and some properties of nZVI were listed in Table 1.

$$4Fe^{2+}{}_{(aq)} + BH_4{}^{-}{}_{(aq)} + 3H_2O \rightarrow 3Fe^0 + H_3BO_{3(aq)} + 7H^{+}{}_{(aq)} \quad (1)$$

$$Fe(C_2H_3O_2)_2 + C \rightarrow Fe^0 + 2CH_2CO + CO + H_2O \quad (2)$$

$$Fe_3O_4 + 2C \rightarrow 3Fe^0 + 2CO_2 \quad (3)$$

2.2. Bimetallic nanoparticles: doping with transition metals

Fresh nZVI easily agglomerates and to be oxidized, which greatly decreases its reactivity. Furthermore, under anaerobic conditions, the reaction of nZVI and water produces hydrogen with high binding energy and unstable electrons, which cannot be used directly to process organic pollutants. To enhance the utilization efficiency of hydrogen and electrons, some transition metals such as Pd, Pt, Ag, Cu and Ni were doped into nZVI (Grittini et al., 1995; Zhang et al., 1998; Xu and Zhang, 2000). One well-known degradation process for organic pollutants by these metal-doped nZVI was dehalogenation, which was dominated

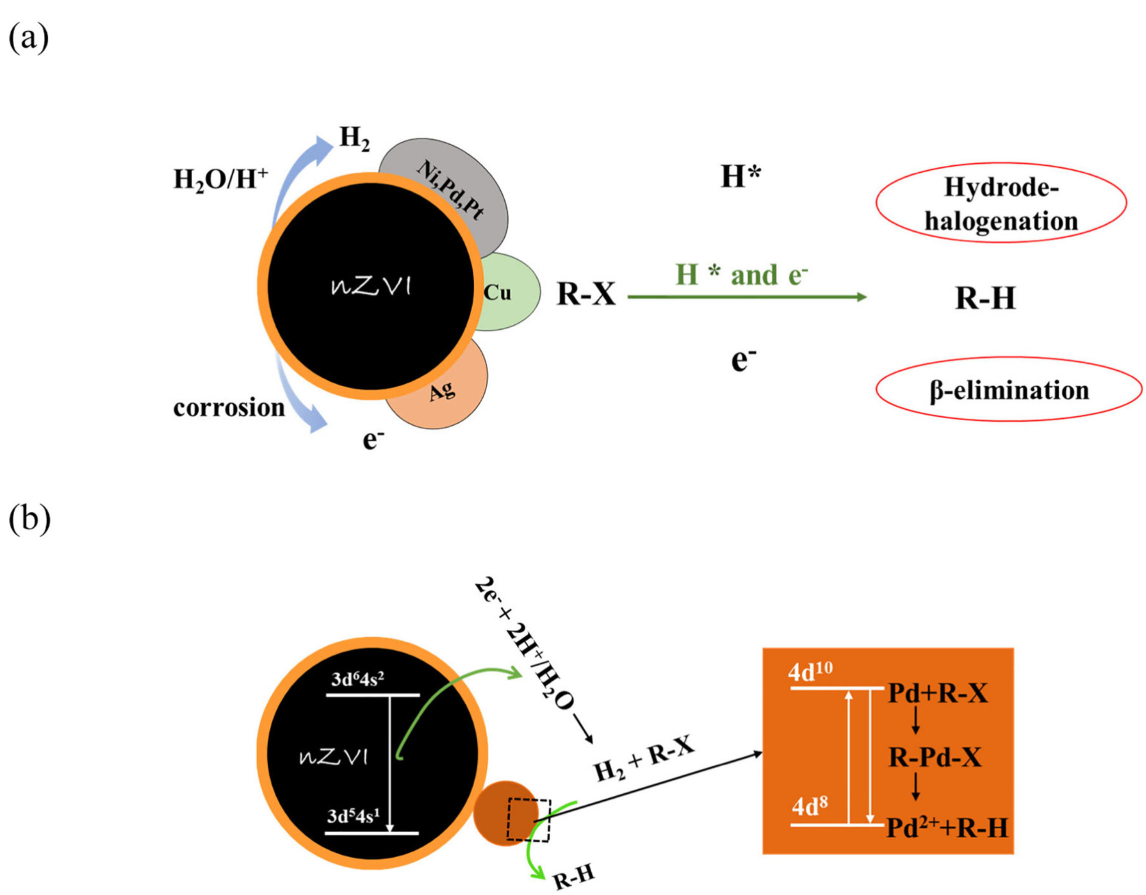

Fig. 1. (a) Mechanisms of dehalogenation by nZVI doped with noble metals; (b) The electron transfer and specific degradation steps of organohalides by Pd/Fe.

by an H-transfer dominant mechanism and proceeded via a stepwise reaction assisted by the hydrogenation catalyst. For example, Pd decomposed hydrogen gas to atomic hydrogen, which adsorbed on the surface of Fe/Pd and contributed to the hydrodehalogenation of organohalides (He and Zhao, 2008). This H-transfer dominant mechanism was also suitable for Ni and Pt (Lin et al., 2004; Bokare et al., 2008; Fang et al., 2011; Dong et al., 2018). The catalytic efficiency of Ni was lower than Pd, but the affordability of Ni promised a greater chance of widespread adoption.

$$Fe + 2H_2O \rightarrow Fe^{2+} + 2OH^- + H_2 \uparrow \text{ neutral and alkaline environment} \quad (4)$$

$$Fe + 2H^+ \rightarrow Fe^{2+} + H_2 \uparrow \text{ acid environment} \quad (5)$$

The other predominant mechanism in organic pollutants transformation by these metal-doped nZVI was direct e^--transfer, which was supported by Ag with direct current flow from Fe to Ag and further electron transfer to organic pollutants (Wang et al., 2015). Due to different catalytic mechanisms, Pd and Ag have selectivity for different organic pollutants corresponding to different degradation mechanisms. Different from other metals, both H-transfer and e^--transfer were discovered on the Cu surface when Cu was applied in combination with Fe (Zhu et al., 2010; Wang et al., 2019). The catalytic mechanisms of these metals for degrading organohalides were illustrated in Fig. 1. When considering groundwater remediation, even with the high reactivity of bimetallic NPs, the high cost still limited their application. It should be emphasized that soil is a more complex matrix than water, and the availability of bimetallic NPs is not as good as in water.

2.3. Stabilized nZVI: coating with macromolecules or surfactants

Although the electron utilization efficiency of nZVI has been enhanced by doping metals, agglomeration was still a challenge because of interparticle interactions, including van der Waals forces, magnetic dipolar interactions, and electric dipolar interactions. To stabilize nZVI, great efforts have been devoted to using carboxymethyl cellulose (CMC), surfactants, etc. (Berge and Ramsburg, 2009; He et al., 2010). The stabilization of nZVI can be carried out in two synthetic methods, which were done either before agglomeration (pre-agglomeration stabilization) or after agglomeration (post-agglomeration stabilization) (Phenrat et al., 2008). In pre-agglomeration stabilization, stabilizers were added during the growth of nanoparticles or to fresh nZVI. For example, CMC has been used to stabilize nZVI and Fe/Pd nanoparticles for TCE dechlorination. During pre-agglomeration stabilization, CMC accelerated the nucleation of Fe atoms and then formed a bulky negatively charged layer around nZVI, preventing agglomeration. In post-agglomeration stabilization, ultrasonic dispersion was used on agglomerated nanoparticles during production or before stabilizers were added. Comparatively, the pre-agglomeration stabilization generated smaller nanoparticles with higher organic pollutant-degrading activity. In particular, the stabilizer-to-nZVI ratio was found to be an important factor for the degrading activity of stabilized nZVI. The minimum stabilizer needed to fully stabilize quantitative nanoparticles was called the critical stabilization concentration (CSC) (Liang et al., 2012). Higher stabilizer concentrations than the CSC help to preserve the nanoparticles' reactivity by producing even smaller particles, but too high stabilizer concentrations decrease the nanoparticles' reactivity by covering the reactive sites.

Apart from the concentrations of stabilizers, different properties of stabilizers also affect the reactivity of nZVI. Anionic, cationic, and nonionic surfactants were used to modify nZVI surfaces and were further applied to the sorption, dissolution, and desorption of organic pollutants to materials or water (Zheng et al., 2009; Tian et al., 2020). Anionic surfactants such as sodium dodecylbenzene sulfonate (SDBS) enhanced the negative potential of the nZVI surface by providing sulfonated head groups. Nonionic surfactants and biosurfactants inhibited the agglomeration of nZVI and increased mobility by coating nZVI (Basnet et al., 2013; Soukupova et al., 2015). Other materials used to stabilize nZVI included gum karaya, xanthan gum, activated carbon, silica and so on (Zhan et al., 2008; Comba and Sethi, 2009). As a natural hydrocolloid gum, gum karaya stabilized nZVI, yielding superior stability by forming a scaffold to prevent nZVI aggregation (Vinod et al., 2017). Active materials were used to improve organic pollutant degradation by forming a new iron carbide phase with high adsorption capacity for organic pollutants (Cheng et al., 2020; Qian et al., 2020). Active materials were also used to stabilize bimetallic NPs that could substantially reduce hydrogen production (Wang et al., 2018). Different coated models of nZVI by CMC, surfactant and gum karaya through surface modification and/or creating a network were shown in Fig. 2. In the surface modification, stabilizers were attached to the nZVI surface to enhance repulsive forces; they also formed a viscous matrix through macromolecule entanglements.

2.4. Sulfidation of nZVI

Sulfidated nZVI (S-nZVI) has been extensively studied for enhancing the activity of nZVI for economic reasons in recent years (Rajajayavel and Ghoshal, 2015; Guan et al., 2019). The synthesis methods of S-nZVI with $NaBH_4$, $FeSO_4 \cdot 7H_2O$ and Na_2S (or $Na_2S_2O_4$) as substrates included "one-pot approach" and "two-step approach". The coprecipitation of iron sulfides and Fe^0 was defined as a "one-pot approach" in which $NaBH_4$, $FeSO_4 \cdot 7H_2O$ and Na_2S (or $Na_2S_2O_4$) were added together; sulfurizing the already formed nZVI was called a "two-step approach" (He et al., 2018). Analysis showed that nZVI hydrophobicity was increased by sulfidation and therefore inhibited the reaction of nZVI with water, resulting in less hydrogen evolution and extended longevity (Li et al., 2017). Electron transport was also enhanced by iron sulfides existing on the S-nZVI surface, resulting in increased electron utilization efficiency (Xu et al., 2019a).

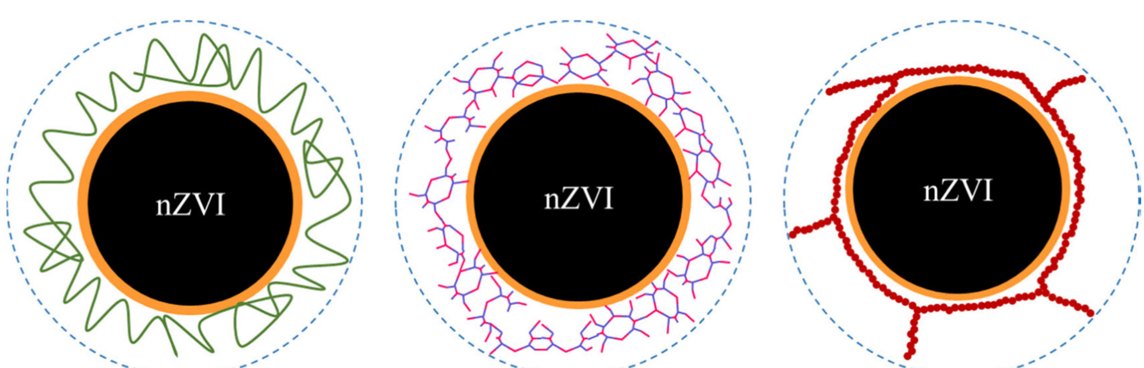

Fig. 2. Stabilized models of nZVI from left to right: CMC, gum karaya and surfactant.

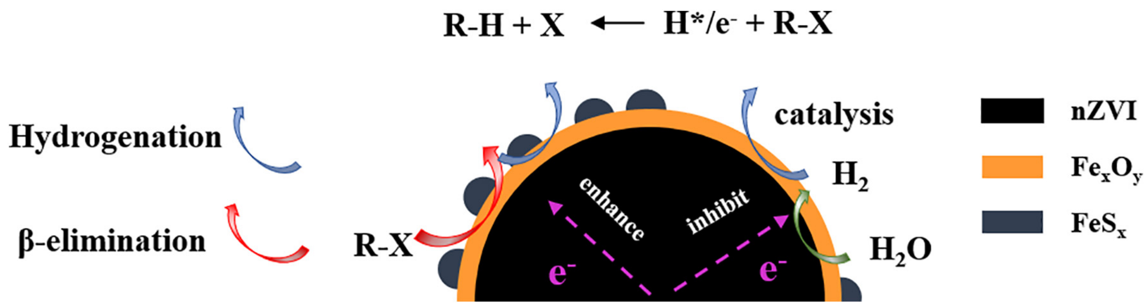

Fig. 3. Dehalogenation mechanisms of organic pollutants by sulfidated nZVI.

The sulfur content of S-nZVI was an essential control on the reactivity of nZVI. Low sulfur content (0.02–0.65 mol% S/Fe) inhibited hydrogen evolution of S-nZVI (up to 13-fold) and increased electron utilization efficiency (up to 20-fold), thereby increasing TCE degradation efficiency by up to 14-fold (Xu et al., 2019a). When the S/Fe molar ratio was ≥0.3, the reaction of nZVI with water was eliminated (Fan et al., 2016). With a proper sulfur content, stable particle structures such as nanocrystalline mackinawite were formed; these cannot be oxidized and merely transfer electrons from the Fe^0 core to organic pollutants. The enhanced electron transport efficiency on the S-nZVI surface resulted in a higher diversity of degradation products than nZVI. The degradation product of TCE by nZVI was ethene, while the degradation products of TCE by S-nZVI were separated into ethane, ethene and acetylene (Xu et al., 2019b). The diversity of TCE degradation products indicated that multiple dechlorinating mechanisms coexist via hydrogenation and β-elimination (Fig. 3). In addition, some metal ions of environmental concern had an effect on the reactivity of S-nZVI. As reported by Kim et al. (2014a), Mn^{2+} and Cu^{2+} showed negative effects, while Pd^{2+}, Co^{2+}, and Ni^{2+} showed positive effects. Hence, it is necessary to consider the combined environmental influences of S-nZVI to obtain the highest remediation efficiency of contaminated soil.

From the view of chemical reactions, carboxylation of nZVI with carboxylic acids such as oxalic acid (OA), citric acid (CA) and formic acid (FA) can promote nZVI corrosion by providing H^+ and transform the nZVI surface properties by providing carboxyl functional groups. In the presence of pH buffering carboxylic acids (i.e., 30 mM FA), the TCE removal rate by nZVI increased 1.48-fold because the optimal TCE degradation pH of ~4.9 was maintained; when the pH was lower than 4.9, the faster disappearance of nZVI decreased the TCE removal rate (Chen et al., 2001). OA and CA have a strong tendency to form complexes with ferrous ions or ferric ions and can degrade a limited set of organic pollutants, such as pentachlorophenol (Hou et al., 2009). These complexes also covered the nZVI surface and inhibited the adsorption of organic pollutants. Finding the critical concentration of carboxylic acids is important to optimize organic pollutants removal under different conditions.

3. Application of nZVI for organic contaminated soil remediation

nZVI was first employed to remove organic compounds in soil in 1997, and the authors reported the complete removal of TNT (70 mg TNT $liter^{-1}$ spiked with ^{14}C-TNT) through sequential treatment with Fe^0 (5% w/v) followed by H_2O_2 (1% v/v) (Hundal et al., 1997). However, in situ remediation of contaminated soil, especially for persistent organic pollutants, was relatively difficult due to the decreased reactivity and mobility of nZVI. As a consequence, the soil remediation by nZVI assisted with physical, chemical and microbial methods and the related factors influencing performance were summarized in this section.

3.1. Electrokinetics-permeable reactive barrier assisted technology

nZVI technology assisted by electrokinetics is a promising remediation technology. First, more electron sources were available, including direct current, oxidation of nZVI, and electrolysis of water. Second, electric effects (electroosmosis, electron transport, and electrophoresis) derived from the electric field facilitated pollutant accumulation and centralized treatment by nZVI (Moon et al., 2005). Third, the H^+ generated near the anode from water electrolysis (Eqs. (6) and (7)) in the electrokinetic process washed away the precipitates on the surface of nZVI to maintain its activity (Chang and Cheng, 2006; Huang and Cheng, 2012). A permeable reactive barrier (PRB) packed with (n)ZVI coupled to electrokinetics (EK-PRB-nZVI) was applied to remediate PCE-contaminated soil by Chang and Cheng (2006). The PCE-removal percentage reached 90% after a 10-day treatment under optimal conditions. Even for soil contaminated with persistent chlorinated hydrophobic organic compounds (CHOCs), EK-PRB-nZVI technology was still effective. A maximum of 83% PCB removal was obtained by using EK-PRB-nZVI technology in PCB-contaminated soil (Gomes et al., 2015). As estimated, the energy and nZVI costs for a full-scale reactor were 12.3 times lower than the average off-site incineration in Europe. Combined with the updated remediation diagram, we showed two different styles in Fig. 4 (Zhou et al., 2020).

$$\text{Anode}: H_2O \rightarrow 2H^+ + 1/2\,O_2\,(g) + 2e^-\ E_0 = -1.229\ V \tag{6}$$

$$\text{Cathode}: 2H_2O + 2e^- \rightarrow 2OH^- + H_2\,(g)\ E_0 = -0.828\ V \tag{7}$$

In addition to electrokinetic technology, UV light and ultrasound are also used to assist nZVI technologies in the remediation of contaminated soil. In contrast to electrokinetic remediation, nZVI technology assisted with UV light usually took place under aerobic conditions. First, nZVI reacted with O_2 to produce hydroxyl radicals for oxidizing organic pollutants. Second, the combination of UV with nZVI produced Fe^{2+} and H_2O_2, allowing a Fenton-like reaction. Combining UV with Fe^0, the degradation rate constant of 1,4-dioxane greatly increased from $4.8 \times 10^{-4}\ min^{-1}$ (only Fe^0) to $19 \times 10^{-4}\ min^{-1}$ (Son et al., 2009). In contrast, ultrasound physically increased the dispersion of nZVI, allowing smaller nanoparticles to be formed (Jamei et al., 2013). Under high ultrasonic power, the size of half of the nanoparticles decreased from 90.3 nm to 29.9 nm, and the BET surface area increased from 10 m^2/g to 42 m^2/g. When used to remove organic pollutants, the removal percentage of methyl orange and metronidazole by rectorite-supported nZVI was enhanced 1.7 and 1.8-fold, respectively, with ultrasound (Yuan et al., 2016). Unfortunately, the high energy demand and operating noise restricted the widespread application of UV-light and ultrasonic-assisted technologies in soil remediation.

3.2. Limited application of bimetallic NPs and sulfidated nZVI

With increased reactivity and mobility, modified nZVI has better application prospects than their unmodified counterparts. Many researchers have reported the study of bimetallic NPs and S-nZVI at the lab scale, although their application in soil remediation is less prevalent. At the lab scale, the removal of 2,2′,4,5,5′-pentachlorobiphenyl from contaminated soil by Pd/Fe was as high as 76.9% under optimal

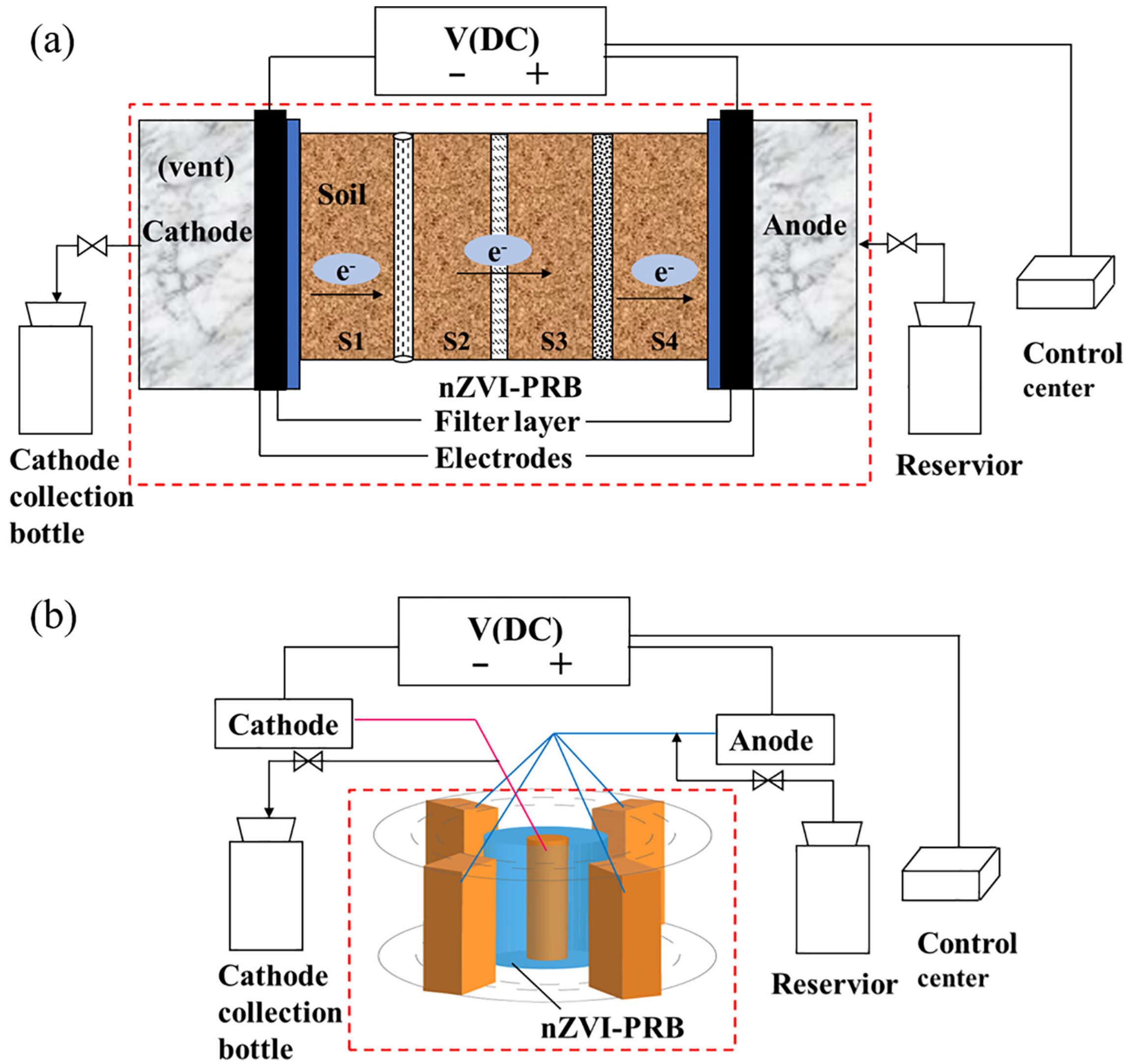

Fig. 4. (a) The common schematic diagram of EK-PRB setup; (b) The new-style schematic diagram of EK-PRB setup.

conditions, including mild acidity, higher Pd loading and higher dosage of Pd/Fe (He et al., 2009). Later, an automatic pH control system was invented to facilitate the remediation of PCB-contaminated soil by nZVI and Pd/Fe (Wang et al., 2012). Due to the high cost and other potential problems, the large-scale application of bimetallic NPs was limited. First, the oxidation and agglomeration of bimetallic NPs decreased the mobility of bimetallic NPs and the removal of organic pollutants. Especially for soil, complex compositions such as organic matter and inorganic minerals interacted with nZVI and hindered the migration of nZVI. Second, bimetallic NPs were difficult to recycle and had shown cytotoxicity to environmental microorganisms. Four types of bimetallic NPs (Fe/Pd, Fe/Cu, Fe/Ni, and Fe/Pt) exhibited varying degrees of cytotoxicity towards *Escherichia coli*, of which Fe/Pd was the least toxic (Kim et al., 2014b). Membrane disruption and cell leakage induced by direct contact with bimetallic NPs were possible mechanisms of cell death.

Different from bimetallic NPs, the application of less expensive S-nZVI was shown to have higher electron utilization efficiency than nZVI. Under aerobic conditions, S-nZVI served as an effective Fenton-like catalyst for oxidizing a variety of organic pollutants (Dong et al., 2019). Under anaerobic conditions, S-nZVI was used as the primary reductant to degrade organic pollutants, especially for chlorinated aliphatic compounds (Xu et al., 2020). It was interesting to note that S-nZVI displayed distinct properties from pure nZVI. The weak acid solution was beneficial for the remediation of TCE- or PCB-contaminated soil using nZVI, and TCE removal using nZVI was not observed at pH values of 9 and 10 (Chen et al., 2001). Nevertheless, the TCE removal rate constant using S-nZVI increased from 0.104 h^{-1} to 0.240 h^{-1} as the pH increased from 7 to 11 due to the deprotonation of FeS increasing the electron availability (Rajajayavel and Ghoshal, 2015).

In regard to real soil remediation, the longevity and mobility of remediation materials must be considered. In a S-nZVI treatability study for 1,2-dichloroethane, 63% of iron was still in the zero-valent state after 400 days, proving the higher corrosion-resistance with extended longevity of S-nZVI than nZVI (Garcia et al., 2016; Vogel et al., 2019). Furthermore, the mobility of S-nZVI was higher than that of nZVI in saturated soil columns due to decreased electrostatic interactions between sand and nanoparticles (Su et al., 2020). Thus, the agglomeration of nZVI-nZVI and nZVI-sand both decreased by sulfidation. As the process of sulfidation is part of the natural biogeochemical cycle, it presents a great applied prospect. In addition, the coexistence of sulfate-reducing bacteria and organohalide-respiring bacteria might enhance the removal percentage of organohalides through competition and synergy. At present, the environmental applications of S-nZVI for soil remediation contaminated with organic compounds are still an actively evolving area.

3.3. Extensive application of stabilized nZVI

Rather than bimetallic NPs and S-nZVI, stabilized nZVI has been more widely applied for remediating soil contaminated with organic compounds due to its higher reactivity, stability and mobility. At the lab scale, He and Zhao (2005) used water-soluble starch to stabilize nZVI, which was able to remove 98% of TCE within 1 h and 80% of PCBs within 100 h. To further enhance the reactivity of nZVI for remediating contaminated soils, CMC-stabilized Pd/Fe particles were used to remove TCE, and the removal percentage of TCE from contaminated soils by CMC-stabilized Pd/Fe particles was enhanced 2-fold higher than the starch-stabilized counterparts (He et al., 2007). In addition, soil properties had an effect on the remediation, which was strongly limited by desorption kinetics of organic pollutants; for example, ~44% of TCE sorbed to potting soil and ~82% of TCE sorbed to Smith Farm soil was removed in 30 h (organic matter content was 0.7%) (Zhang et al., 2011). Obviously, the activity of stabilized nZVI or bimetallic NPs was higher than that of pure nZVI or bimetallic NPs in anaerobic/aerobic processes. The biocidal effects of stabilized nZVI were also lower than pure nZVI, a long-term positive impact of CMC-stabilized nZVI on in situ microbial communities has been detected (Kocur et al., 2016). We described the application of nZVI- and CMC-stabilized nZVI in Fig. 5. When injected into contaminated sites, the mobility of stabilized nZVI was also enhanced compared to pure nZVI. Surface coatings protected against nZVI agglomeration and produced smaller nanoparticles, which fit into the interlayer spaces between soil particles and traveled longer distances before being trapped by the soil matrix. The transport of stabilized nZVI has been simulated using classical filtration theory, and the retention of nanoparticles in porous media mainly included three mechanisms: direct interaction of the nanoparticles with the media, sedimentation, and diffusion (Yao et al., 1971; He et al., 2009). The model simulation results indicated that 99% of CMC-stabilized nZVI would be removed by the soil matrix within 16 cm under groundwater flow (0.1 m/day) but may travel over 146 m under pore flow (61 m/day). This proved that the particles' travel distance was strongly dependent on interstitial flow velocity. In recent years, CMC-stabilized and sulfidated nZVI was applied to in situ treatment contaminated with organic pollutants. In a field study, the reactivity and mobility of nZVI were both enhanced following short-term abiotic and long-term biotic dechlorination (Garcia et al., 2020). It is likely to achieve a balance between the stability and longevity/capacity of nZVI. We have compared the stability and mobility of different modified nZVI from previous studies in Table 2.

As another important stabilizer for nZVI, surfactants increased the reactivity and mobility of nZVI as well as the solubility of organic pollutants in particular. Under optimal conditions, the TCE removal by cationic hexadecyl trimethyl ammonium bromide (CTAB)- or anionic sodium dodecyl sulfate (SDS)-modified nZVI reached 100% after 4 h by dispersing nZVI and solubilizing TCE in a soil-water system (Tian et al., 2018). Further optimized remediation methods using saponin- or Tween 80-stabilized nZVI assisted by electrodialysis enhanced the removal of organic compounds (Suanon et al., 2020). Different from CMC-stabilized nZVI, surfactant-stabilized nZVI had a lower reactivity towards TCE and higher mobility in situ due to emulsification (Basnet et al., 2013). At present, there were two critical issues in the application of surfactants: finding the optimal concentration and mediating microbial toxicity. First, the high activity and effective stabilization were difficult to balance. The stabilization of nZVI was enhanced with increasing concentrations of surfactants, while the reactivity remained stable. Excessive surfactants were likely to occupy the adsorption sites on the nZVI surface and further decrease the removal of organic pollutants. Second, some kinds of surfactants might be toxic to indigenous microorganisms. We have summarized the applications of chemically modified nZVI for soil remediation in Table 3.

Different from the above reduction technology, advanced oxidation technology leveraging iron-based nanomaterials has also been used for soil remediation to achieve the mineralization of organic pollutants (Bergendahl and Thies, 2004; Zhao et al., 2010). The most common advanced oxidation technology was activating persulfate (PS) with nZVI or S-nZVI. In the Fe^0/PS system, rapid TCE mineralization was achieved accompanied by rapid persulfate decomposition and chloride ion formation (Liang and Lai, 2008). In spiked and aged soil, the removal percentage of PCBs was as high as 70.0% after 3 days using an Fe^0/PS system (Tang et al., 2015). The S-nZVI has been used to activate PS with better pH adaptiveness and higher removal percentage of organic pollutants than nZVI (Dong et al., 2019). The TCE removal increased from 43.8% to 76.5% by substituting nZVI with S-nZVI at pH 6.5. Obviously, advanced oxidation technology supported iron-based materials were highly efficient and rapid for soil remediation, but it greatly perturbs the soil ecosystem.

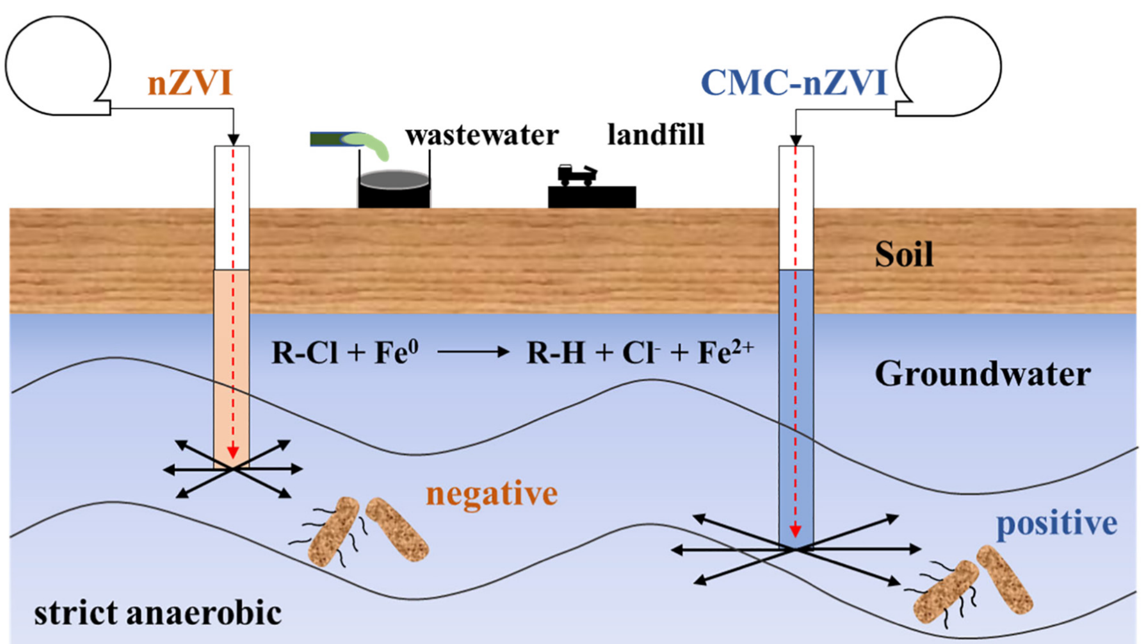

Fig. 5. Application of nZVI and CMC stabilized nZVI for organohalides removal.

Table 2
The stability and mobility of different modified nZVIs.

Form of nZVI	Contaminated sites	Stability	Mobility	References
CMC stabilized nZVI	Contaminated sandy subsurface area in Sarnia, Ontario	There was no significant aggregation of stabilized nZVI in site-synthesis compared with previous work.	The breakthrough of normalized total Fe increased from 37% to 75% at 1 m away from the injection well.	He et al., 2010; Kocur et al., 2014
Sulfidated nZVI	Lab-scale	63% of the iron was still in the zerovalent state after 400 days.	/	Garcia et al., 2016
Sulfidated ZVI/AC	Lab-scale	The corrosion rate of nZVI was decreased 65-fold.	/	Vogel et al., 2019
CMC stabilized and sulfidated nZVI	Injection well in Sarnia, Ontario	Up to 17 days, significant amounts of total iron were found in the farthest monitoring well.	The maximum delivery distance was 2.7 m for the NA4-Blue well which was the site of the farthest monitoring well.	Garcia et al., 2020
Sulfidated nZVI	Different water bodies and the sand column	The amount of Fe in the effluent increased by 16–37% upon the sulfidation of nZVI.	The mobility of two nanoparticles was S-nZVI > nZVI.	Su et al., 2020

3.4. nZVI combination with microbial technology

Due to its broad applicability and environmental friendliness, the application of integrated nZVI with microbes, especially for organohalide-respiring bacteria, has been extended to numerous organic pollutants with several advantages. First, organohalide-respiring bacteria needed hydrogen as their electron donor to obtain energy for growth during the dehalogenation of organohalides, which could be provided by the corrosion of nZVI or other anaerobic microbes (Seshadri et al., 2005; Li et al., 2019). Lampron et al. (1998) first reported that nZVI improved the biological reduction of TCE by providing electron donors. Second, nZVI created favorable conditions for soil bioremediation by organohalide-respiring bacteria. nZVI decreased the redox potential of the dehalogenation environment, which shortened the dehalogenation lag phase of organohalide-respiring bacteria and reduced the stability of organohalides (Holmes and Harrison, 1993; Shi et al., 2010). In the study of Zanaroli et al. (2012), nZVI (6.7 g/kg) decreased the lag phase of PCB microbial dechlorination from 30 weeks to 20 weeks in marine sediment due to the low redox potential. Third, nZVI and microbes can remove organohalides by attacking halogens at distinct and specific positions of the organohalides to achieve complete dehalogenation. nZVI could rapidly detoxify highly chlorinated aliphatic hydrocarbons (CAHs) but was unable to remove low CAHs, while low CAHs could be removed by organohalide-respiring bacteria (Koenig et al., 2016). Fourth, the electron utilization efficiency of nZVI was enhanced by organohalide-respiring bacteria by employing several sets of functional enzymes, including hydrogenases and reductive dehalogenases (RDase) (Rosenthal et al., 2004; Rahm et al., 2006). To date, five putative hydrogenase genes of organohalide-respiring bacteria have been identified. The related RDase genes of organohalide-respiring bacteria were listed in Table 4 (Rahm et al., 2006; Türkowsky et al., 2018). These functional enzymes facilitated electron transfer from nZVI to organohalides (Fig. 6). Combined abiotic and biotic methods can completely remove organohalides to nontoxic products with high efficiency. In addition, the interactions of nZVI with other soil microbes have also been considered. For example, *Shewanella alga* BrY, a dissimilatory iron-reducing bacterium, stimulated carbon tetrachloride transformation by nZVI and reduced the oxidized nZVI surface following increased ferrous iron (Gerlach et al., 2000). An integrated treatment with nZVI and microbes effectively removed tetracycline and increased the proportions of *Comamonadaceae*, *Oxalobacteraceae* and *Chromatiaceae* (Huang et al., 2017). Based on these results, the application of integrated microbes and nZVI treatments for organohalide-contaminated soil remediation is promising.

Despite their promising synergies, several challenges were inevitable in the integrated nZVI and microbe system. First, oxidation and agglomeration lowered the reactivity and mobility of nZVI, forming a gap between nZVI and organohalide-respiring bacteria. The period of abiotic dehalogenation mediated by nZVI was hours to days, while microbial reductive dehalogenation was extremely slow, occurring on the scale of months (Nurmi et al., 2005; Löffler et al., 2013). Second, the cytotoxicity of nZVI was evident, which included disturbance of the cell membrane, DNA/protein damage and down-regulation of gene expression (Xiu et al., 2010; Chaithawiwat et al., 2016). The integrity and permeability of the cell membrane was destroyed by the physical attachment of nZVI to *Escherichia coli*. The phospholipids and proteins on the cell membrane of *E. coli* might be damaged by nZVI through the generation of Fe^{2+} and reactive oxygen species (Lee et al., 2008). The expression of RDase genes in *Dehalococcoides* was down-regulated by bare nZVI, while the expression of RDase genes was up-

Table 3
The removal of organohalides by nZVI in soil.

Target groundwater/soil	Treatments	Main performances	References
PCB-contaminated soil	Fe^0 and Pd/Fe^0	The dechlorination rate constants were enhanced 1.24–2.69 fold with Pd.	Wang et al., 2012
DDT-contaminated soil	nZVI	1 g nZVI kg^{-1} was efficient for significant DDT degradation while a higher concentration was necessary for treating aged pollutants in soil.	El-Temsah and Joner, 2013
PCBs-contaminated soil	nZVI and surfactant	Surfactants enhanced PCB desorption and dechlorination in contaminated soil.	Gomes et al., 2014
TCE-contaminated soil	CMC stabilized nZVI	TCE concentrations in groundwater decreased by more than 99%.	Chowdhury et al., 2015
PCB-contaminated soil	nZVI and electric current	83% of PCB was removed and the cost at 72 € for each cubic meter of PCB contaminated soil was low.	Gomes et al., 2015
PAHs-contaminated soil	Iron-activated persulfate	nZVI were more efficient to activate persulfate than Fe^{3+} for degrading PAHs.	Pardo et al., 2016
2,3,7,8-TCDD contaminated soil	CMC-coated nZVI and surfactant	The maximum removal efficiency was approximately 60% even at an extremely low pH (approx. 3.6).	Binh et al., 2016
DDT-contaminated soil	nZVI and sodium oleate -nZVI	The degradation halflife decreased from 58.3 to 27.6 h with nZVI dosage from 0.5 to 2.0% and from 46.5 to 32.0 h with temperature from 15 to 35 °C.	Han et al., 2016
PCBs-contaminated soil and water	nZVI and mZVI	All types of nZVI and mZVI were highly efficient in degradation of PCBs in water, but had little degradation effect on PCBs in soil.	Ševců et al., 2017
TCE-contaminated soil	PEG-4000-nZVI and surfactant	The removal efficiency of TCE in a soil-water system reached 100% after 4 h at specific experimental conditions.	Tian et al., 2018

Table 4
Overview of RDase genes for the degradation of organohalides.

Microorganisms	Genes	Substrates	End products	References
Dehalococcoides mccartyi 195	*tceA/pceA*	TCE/PCE	VC/TCE	Seshadri et al., 2005
Dehalococcoides mccartyi FL2	*tceA*	TCE	VC	He and Zhao, 2005
Dehalococcoides mccartyi 11a5/11a	*tceA/pteA/vcrA*	TCE/PCE/VC	VC/PCE/ETH	Lee et al., 2013
Dehalococcoides mccartyi BTF08	*tceA/vcrA/pceA*	TCE/VC/PCE	VC/ETH/TCE	Pöritz et al., 2013
Dehalococcoides mccartyi ANAS1/ANAS2	*tceA/vcrA*	TCE/VC	VC/ETH	Lee et al., 2011
Dehalococcoides mccartyi AD14-1/AD14-2	*tceA/vcrA*	TCE/VC	VC/ETH	Wang and He, 2013
Dehalococcoides mccartyi UCH007	*tceA/vcrA*	TCE/VC	VC/ETH	Uchino et al., 2015
Dehalococcoides mccartyi VS/BAV1	*vcrA/bvcA*	VC	ETH	Waller et al., 2005
Dehalococcoides mccartyi GT	*vcrA*	VC	ETH	Sung et al., 2006
Dehalococcoides mccartyi 11G	*vcrA*	VC	ETH	Zhao and He, 2015
Dehalococcoides mccartyi MB	*mbrA*	TCE	DCE	Low et al., 2015
Dehalococcoides mccartyi CG1/CG4/CG5	*pcbA1/pcbA4/pcbA5*	PCE/PCBs	DCE/PCBs	Wang et al., 2014
Dehalococcoides mccartyi CBDB1	*cbrA*	CBs/PCE	CBs/DCE	Adrian et al., 2007
Dehalococcoides mccartyi DCMB5	*cbrA*	CBs/PCE	CBs/PCE	Pöritz et al., 2013
Dehalobacter restrictus PER-K23	*pceA*	PCE/TCE	DCE	Kruse et al., 2013
Desulfitobacterium hafniense Y51	*pceA*	PCE/TCE	DCE	Nonaka et al., 2006
Desulfitobacterium hafniense PCE-S	*pceA*	PCE/TCE	DCE	Miller et al., 1998
Dehalobacter sp. UNSWDHB	*TmrA*	TCM/TCA/DCA	TCA/DCA/VC	Wong et al., 2016
Dehalobacter sp. CF/DCA	*CfrA/DcrA*	CF/TCA/DCA	DCM/DCA/CA	Tang et al., 2016

Note: PCE, tetrachloroethene; TCE, trichloroethene; DCE, dichloroethene; VC, vinyl chloride; ETH, ethene; PCBs, polychlorinated biphenyls; CBs, chlorobenzene; DCA, dichloroethane; TCM, trichloromethane; TCA, trichloroethane; CF, chloroform; CA, chloroethane.

regulated by coated nZVI (Xiu et al., 2010). In fact, the cytotoxicity of nZVI was highly dependent on the dose, and higher nZVI concentrations had more serious inhibitory effects on bacterial communities (Velimirovic et al., 2015). To reduce the corrosion rate and cytotoxicity of nZVI, macromolecules were used to coat nZVI. CMC-stabilized nZVI was injected into a contaminated site to enhance organic compounds removal by organohalide-respiring bacteria, which increased the abundance of *Dehalococcoides* and *Dehalogenimonas* up to 1 order of magnitude (Kocur et al., 2016). In a field test, carbo-iron particles offered additional possibilities to support microbial dechlorination, such as sorption with its component structure (Vogel et al., 2019). These results proved that nZVI had an effect on complex microbial communities from natural and engineered environments.

4. Factors influencing soil remediation

The physical and chemical properties of soils, such as pH, temperature and organic matter, affected the behavior of nZVI. With increased pH, large amounts of H^+ were consumed, which hindered organic pollutants removal by nZVI. Chen et al. (2001) compared the removal of TCE by nZVI at pH values from 1.7 to 10. The optimal removal of TCE was achieved at pH 4.9, and TCE removal was not observed at pH values of 9 and 10. The optimal pH value at approximately 4.90–5.10 was demonstrated by Wang et al. (2012). The normalized pseudo-first-order removal rate constant increased 3-fold for 2,4,4′-CB by nZVI with an automatic pH control system. However, organohalide-respiring bacteria were inactivated in strongly acidic and basic conditions, which hindered the combined efficiency of nZVI and microbial technology (Yang et al., 2017).

Temperature also has an important effect on nZVI technology in soil remediation. From the viewpoint of thermodynamics, increasing temperature enhanced the removal of organic pollutants by nZVI due to the high energy cost. For example, the removal of 2,4-DCP by Pd/Fe increased from 39.1 ± 1.9% to 89.7 ± 0.7% in 5 h as the temperature increased from 283.15 K to 313.15 K (Xu et al., 2013). Unfortunately, the adverse effects of increasing temperature on the activity of nZVI have also been noticed in soil remediation. The removal percentage of 1,2,3,4,5,6-hexachlorocyclohexane (γ-HCH) by nZVI decreased from 99.0% to 78.0% as the temperature increased from 303 K to 308 K, which was caused by the decreased activity of soil microbes (Singh et al., 2013). These results indicated that an optimal temperature is needed for the application of nZVI in practical soil remediation.

Another key factor is natural organic matter (NOM), which controls the corrosion and mobility of nZVI through adsorption, solubility and moisture interactions. Many studies take extracted humic substances

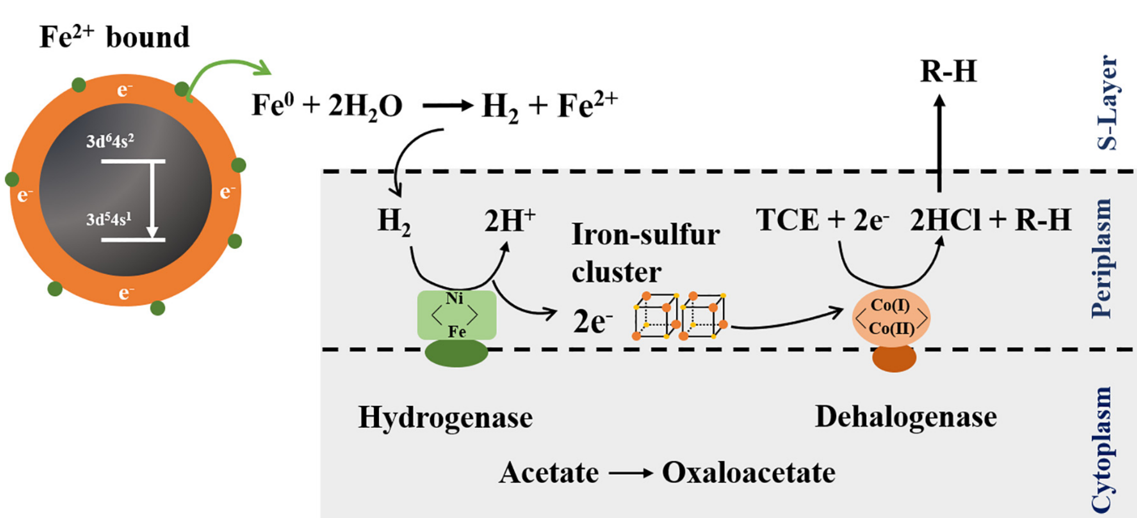

Fig. 6. The mechanism of nZVI integrated with organohalide-respiring bacteria.

such as humic acid (HA) as a representative to investigate the impact of NOM (Han et al., 2019; Yao et al., 2019). HA played an important positive role in the removal of chloramphenicol (CAP), with the removal percentage of CAP by nZVI at 80.0% or higher (Yao et al., 2019). The mass ratio of nZVI/HA and the dosage of nZVI/HA-30 had a great effect on the removal of CAP. Moreover, HA enhanced TCE removal by organohalide-respiring bacteria as an electron transfer medium. The cytotoxicity of nZVI was decreased, and the mobility of nZVI was enhanced by coating organic matter, possibly by reducing the sticking coefficient (Johnson et al., 2009). Inevitably, the negative effect of organic matter on pollutants degradation by nZVI occurred occasionally when excessive HA occupied the active surface sites of nZVI and inhibited contact between nZVI and pollutants (Xu et al., 2013). Last, the interaction of nZVI with inorganic clay mineral gel and humic acid led to phase-out reunion in soil pores, which would significantly change the properties of nZVI, including the size distribution, electrophoretic mobility, surface composition and surface charge properties. These changed properties of nZVI would further affect the migration of nZVI in soil pores and its interaction with organic pollutants. There are some other factors such as ionic strength and dissolved oxygen, that affect soil remediation. Due to the complex composition of soil, the detailed interactions between nZVI and soil composition need to be further explored.

5. Challenge and perspective

The application of nZVI for soil remediation contaminated with organic compounds has constantly grown in recent years due to its relatively low cost, low toxicity, high reactivity, and high surface area. Aqueous-phase reduction is most commonly used method for nZVI synthesis. In order to overcome rapid deactivation of fresh nZVI caused by oxidation and agglomeration, chemical modification methods yielding bimetallic NPs, stabilized nZVI, and S-nZVI have been developed. Among these modified nZVI, stabilized nZVI is most favorable for in situ remediation of contaminated soil. In addition, coupling of nZVI with functional microbes (e.g. organohalide-respiring bacteria) is proven more effective and economical. Despite recognized effectiveness of nZVI for treating contaminated soil, some practical challenges limit its scalable applications. The challenges include:

1) Concerns about the comparatively high cost of transition metals and the potential secondary pollution of bimetallic NPs limit their large-scale application. Less expensive materials as alternative promoters for field application will be of great interest from scientific and practical perspectives.
2) The mechanism underlying the behavior of nZVI particles when being injected into contaminated sites needs further study. How organic matters and inorganic minerals affect reactivity and mobility of nZVI remains unclear. A model developed from both laboratory and field tests will help to evaluate and predict the behavior of nZVI precisely and provide accurate guidance for the application of nZVI in soil remediation.
3) The information about long-term impacts of nZVI on soil ecosystems is not yet sufficient. Addition of nZVI inevitably disturbs the soil ecosystem and shapes the interactions between microorganisms and nZVI, but long-term field studies of nZVI are scarce. Long-term studies on in situ field remediation with nZVI will provide more accurate and comprehensive information about how nZVI affects the soil ecosystem.

In summary, organics-contaminated soil remediation using nZVI is a very promising technology, but several practical concerns need to be addressed. Therefore, there is an urgent necessity of developing a more cost-effective, accurate, and environment friendly strategy based on current progress.

Declaration of competing interest

The authors have no affiliation with any organization with a direct or indirect financial interest in the subject matter discussed in the manuscript.

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Appendix A. Supplementary data

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